

Student Name: _____

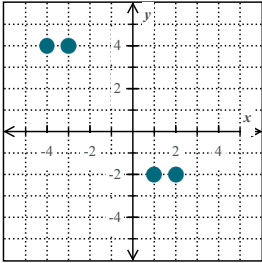
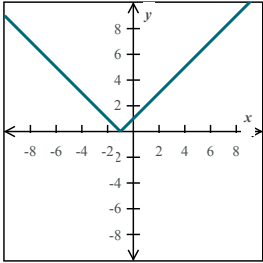
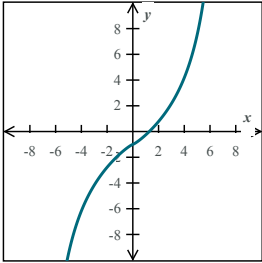
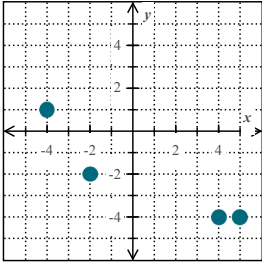
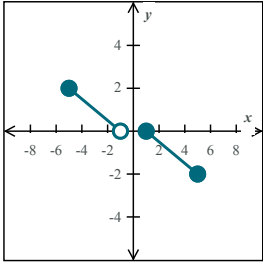
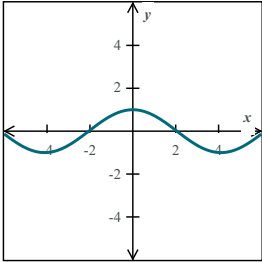
Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **12**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 12

For each function graphed below, state whether it is one-to-one.

	Graph 1 	Graph 2 	Graph 3 
One-to-one?	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No
	Graph 4 	Graph 5 	Graph 6 
One-to-one?	☐ Yes ☐ No	☐ Yes ☐ No	☐ Yes ☐ No

Question 2 of 12

For each pair of functions f and g below, find $f(g(x))$ and $g(f(x))$.
Then, determine whether f and g are inverses of each other.

Simplify your answers as much as possible.
(Assume that your expressions are defined for all x in the domain of the composition.
You do *not* have to indicate the domain.)

(a) $f(x) = x + 3$	(b) $f(x) = -\frac{4}{x}$
$g(x) = x - 3$	$g(x) = \frac{4}{x}$
$f(g(x)) =$	$f(g(x)) =$
$g(f(x)) =$	$g(f(x)) =$
- f and g are inverses of each other	- f and g are inverses of each other
- f and g are <i>not</i> inverses of each other	- f and g are <i>not</i> inverses of each other

(a) $f(x) = x + 3$

$g(x) = x - 3$

(b) $f(x) = -\frac{4}{x}, x \neq 0$

$g(x) = \frac{4}{x}, x \neq 0$

Find the compositions.

Find the compositions.

$f(g(x)) =$ _____

$g(f(x)) =$ _____

$f(g(x)) =$ _____

$g(f(x)) =$ _____

- Are f and g inverses of each other?

☐ f and g are inverses of each other

☐ f and g are *not* inverses of each other
- Are f and g inverses of each other?

☐ f and g are inverses of each other

☐ f and g are *not* inverses of each other

Question 3 of 12

The one-to-one functions g and h are defined as follows.

$g = \{(-7, -2), (-4, 7), (1, 0), (4, -4)\}$

$h(x) = \frac{x - 9}{7}$

Find the following.

$g^{-1}(-4) =$ _____
$h^{-1}(x) =$ _____
$(h \circ h^{-1})(-1) =$ _____

Question 4 of 12

Consider the function $f(x) = \sqrt{2 - x} + 6$ for the domain $(-\infty, 2]$.

Find $f^{-1}(x)$, where f^{-1} is the inverse of f .

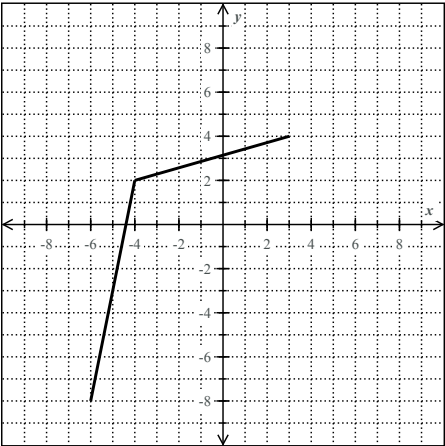
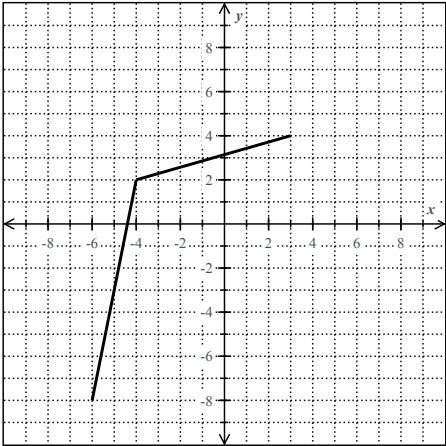
Also state the domain of f^{-1} in interval notation.

$f^{-1}(x) =$ _____ for the domain _____

Question 5 of 12

Below is the entire graph of function f .

Graph f^{-1} , the inverse of f .



Question 6 of 12

The function h is defined by the following rule.

$$h(x)=5^x$$

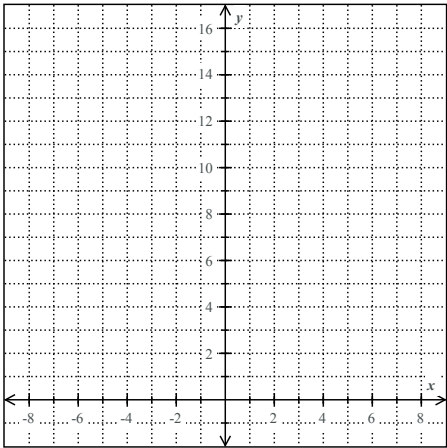
Find $h(x)$ for each x -value in the table.

x	$h(x)$
-1	_____
0	_____
1	_____
2	_____
3	_____

Question 7 of 12

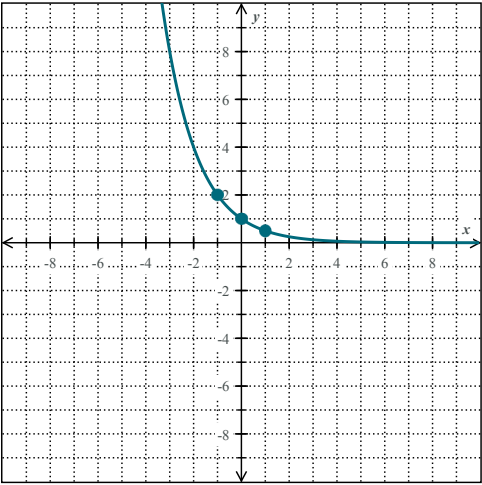
Graph the exponential function.

$$f(x)=\left(\frac{4}{5}\right)^x$$



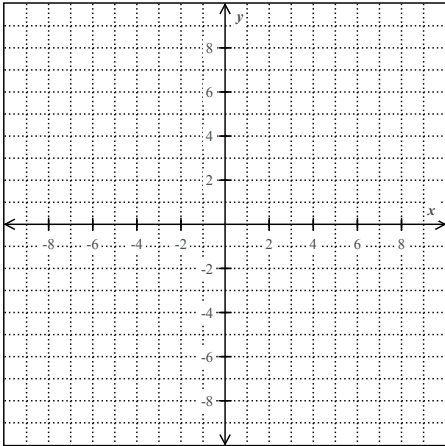
Question 8 of 12

Below is the graph of $y = \left(\frac{1}{2}\right)^x$. Translate it to become the graph of $y = \left(\frac{1}{2}\right)^{x-3} - 1$.



Question 9 of 12

- (a) Graph the exponential function $g(x) = -2^x - 3$.
To do this, plot two points on the graph of the function, and also draw the asymptote. Then click on the graph-a-function button.



- (b) Give the domain and range of the function using interval notation.

Domain: _____

Range: _____

Question 10 of 12

The half-life of a radioactive isotope is the time it takes for a quantity of the isotope to be reduced to half its initial mass. Starting with 140 grams of a radioactive isotope, how much will be left after 6 half-lives? Round your answer to the nearest gram.

 grams

Question 11 of 12

Suppose that \$1400 is borrowed for seven years at an interest rate of 9.5% per year, compounded *continuously*. Find the amount owed, assuming no payments are made until the end.

Do not round any intermediate computations, and round your answer to the nearest cent.

Question 12 of 12

The number of bacteria in a certain population is predicted to increase according to a *continuous exponential growth model*, at a relative rate of 6% per hour. Suppose that a sample culture has an initial population of 87 bacteria. Find the population predicted after two hours, according to the model.

Do not round any intermediate computations, and round your answer to the nearest tenth.

_____ bacteria

Question 1 of 12

	Graph 1	Graph 2	Graph 3
One-to-one?	☐ Yes ☒ No	☐ Yes ☒ No	☒ Yes ☐ No
	Graph 4	Graph 5	Graph 6
One-to-one?	☐ Yes ☒ No	☒ Yes ☐ No	☐ Yes ☒ No

Question 2 of 12

- (a) $f(x) = x + 3$

$g(x) = x - 3$

$f(g(x)) = x$

$g(f(x)) = x$

☒ f and g are inverses of each other

☐ f and g are *not* inverses of each other
- (b) $f(x) = -\frac{4}{x}, x \neq 0$

$g(x) = \frac{4}{x}, x \neq 0$

$f(g(x)) = -x$

$g(f(x)) = -x$

☐ f and g are inverses of each other

☒ f and g are *not* inverses of each other

Question 3 of 12

$$g^{-1}(-4) = 4$$

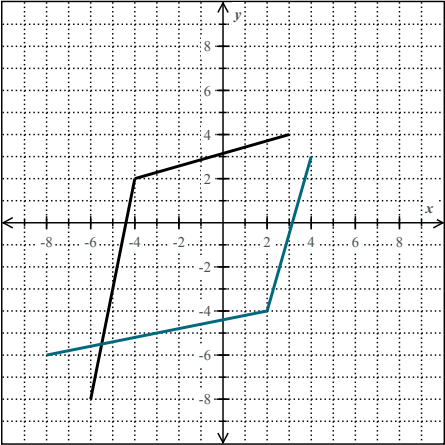
$$h^{-1}(x) = 7x + 9$$

$$(h \circ h^{-1})(-1) = -1$$

Question 4 of 12

$f^{-1}(x) = 2 - (x - 6)^2$ for the domain $[6, \infty)$

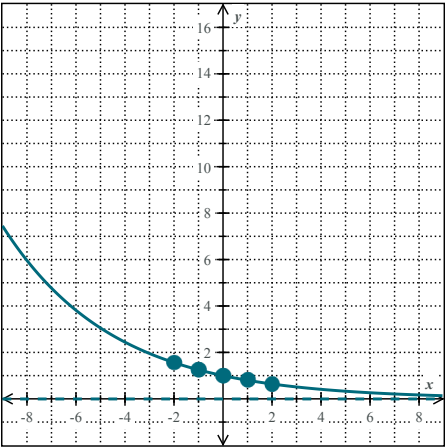
Question 5 of 12



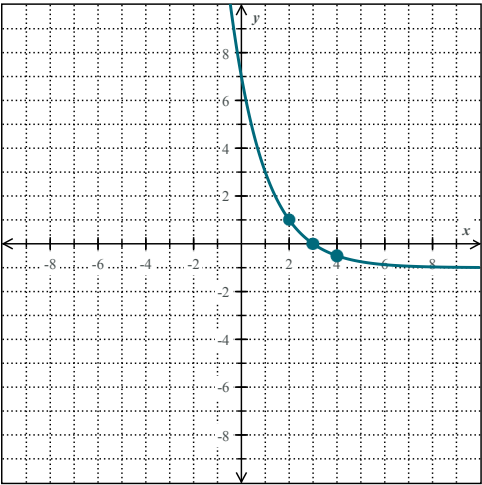
Question 6 of 12

x	$h(x)$
-1	$\frac{1}{5}$
0	1
1	5
2	25
3	125

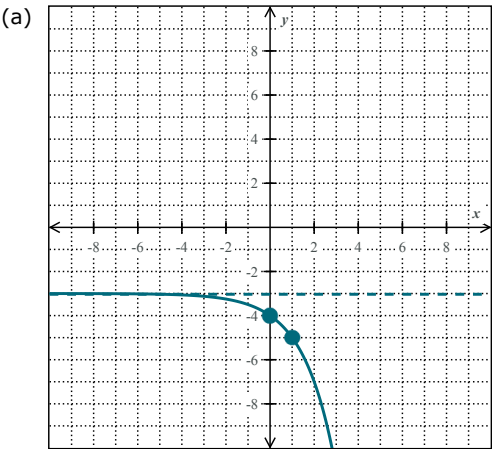
Question 7 of 12



Question 8 of 12



Question 9 of 12



- (b)
- Domain: $(-\infty, \infty)$

Range: $(-\infty, -3)$

Question 10 of 12

2 grams

Question 11 of 12

\$2722.29

Question 12 of 12

98.1 bacteria